SensorGuard

Ground Truth Under Attack

Signed, cross-checked and trust-scored weather stations

Phase 1 technical report · Track 2: False Data Injection, in-situ sensing segment

A prototype that proves where every weather-station reading came from, checks every station against an independent witness, gives each station a calibrated compromise probability and a trust score, and responds in four graded steps, so that a falsified reading does not silently drive an irrigation, drought or insurance decision.

Contents

1 The report in one page **4**

2 The problem **5**

2.1 Why weather stations matter . . . 5
2.2 The three standard attacks . . . 5
2.3 Why stations are easy to attack . . . 5
2.4 Integrity is not truthfulness . . . 6

3 Threat model **6**

3.1 The chain, and where an attacker can stand . . . 6
3.2 What is worth protecting . . . 6
3.3 Catalogue of injection attacks . . . 7
3.4 How a smart attacker might try to dodge . . . 7
3.5 Which layer stops which attack . . . 8
3.6 Residual risk, stated honestly . . . 8
3.7 Is the threat real? . . . 8

4 Architecture and trust anchor **9**

4.1 Six design rules . . . 9
4.2 The whole pipeline . . . 9
4.3 Layer 1: the trust anchor . . . 9
4.4 What a record looks like . . . 10
4.5 How the service checks a record . . . 10
4.6 Layer 1 tested: the same attacks at three positions . . . 11
4.7 What it costs . . . 11

5 Data and signal processing **12**

5.1 The data . . . 12
5.2 Cleaning . . . 12
5.3 A point against a grid cell: calibration . . . 12
5.4 Validation gate: do the residuals behave on clean data? . . . 13
5.5 Replay check and change-point dating . . . 14
5.6 Specified but not built yet . . . 14

6 Detection **14**

6.1 Detector S, the statistical baseline . . . 15
6.2 Detector M, the learned detector . . . 15
6.3 Fusion: one calibrated compromise probability . . . 16
6.4 What is fitted on which data . . . 17
6.5 Day-by-day accuracy and which evidence matters . . . 17

7 Testing against attacks, and the response **18**

7.1 The standard scenario . . . 18
7.2 Results . . . 19
7.3 What decides detection . . . 19
7.4 False alarms, including real extreme weather . . . 20
7.5 Attackers who adapt . . . 20
7.6 The response policy . . . 22
7.7 Farm decisions with and without the defence . . . 23
7.8 Blind rehearsal . . . 24

8 Key metrics at a glance **24**

9 Limits and what comes next **24**

10 Reproducing the results **25**
10.1 File mode: any station series plus coordinates . . . 25
10.2 The control room . . . 25

A Datasets **26**

B Parameters **26**

Glossary **27**

1 The report in one page

The problem. Weather stations in farm fields are treated as the "ground truth". Irrigation advice, drought maps, crop insurance and even the checking of satellite data all rely on them. But a station sits alone in a field, often owned by the person who gains if it reports something false. If someone quietly raises its temperature by 12 %, the farm looks drier, and a decision changes without anyone seeing a forged document.

What we built. A prototype with three layers that fail in different places:

- **Layer 1, provenance.** Every reading is signed by the station itself and chained to the previous one. This stops forgery, changes and replays on the way to the service. It cannot tell if the station itself is lying.
- **Layer 2, an independent witness.** Each reading is compared with NASA POWER (a satellite and model dataset) for the same place and day. The detectors look for disagreement that lasts or looks unusual, not for extreme values.
- **Layer 3, trust and response.** The detector outputs become a probability that the station is compromised, then a trust score, then one of four actions: down-weight, ask for verification, quarantine, alert.

Main results.

- 100 % of 7-day replays were caught on the first day, with zero false alarms from the replay matcher.
- 90–94 % of 12 % steps and 15 % drifts were caught (detector S) when "12 %" means 12 % of the variable's range, and 70–81 % when it means 12 % of the value.
- 4.7 false alarms per station-year, against a CUSUM budget of 1 a year on training years.
- -32 % days with a wrong farm decision when the defence is on (7.1 % $\rightarrow$ 4.9 %).

How it was tested. Three real stations in three climates (Arizona, Florida, Kansas). Everything was set up on 2021–2023 and tested on 2024. The attacker sits at the station and signs the false values itself, which is the hardest case for the signatures. In total 756 attacked runs: three attacks, four variables, both signs, three possible meanings of "12 %", and several start dates and drift lengths.

Other numbers. The compromise probability is well calibrated on stations the model never saw (Brier score 0.131 against 0.162 for a model with no skill). Security costs 118 bytes per record, 62.4 µs to sign and 150.5 µs to verify on a laptop CPU.

The honest limit. Attacks smaller than about 0.7 standard deviations of the station-to-reference difference cannot be seen. They hide inside the normal disagreement between a single station and a 50 km grid cell. This covers every "12 % of the standard deviation" case and most wind cases. The fix is a second independent witness (Sentinel-2 satellite data, neighbouring stations), which is the first planned next step.

2 The problem

2.1 Why weather stations matter

Our "sensor stream" is a network of agricultural weather stations. Each one reports five daily values: temperature, relative humidity, wind speed, rainfall and solar radiation. A service turns them into irrigation advice, drought indicators and index-insurance triggers.

Almost every one of these decisions is a threshold on a water balance: how much rain fell compared with how much water the crop could lose. An index-insurance contract pays when the 30-day balance drops below a trigger. Raise the recorded temperature or solar radiation by 12 % and the computed evapotranspiration goes up, so the field looks drier. Replay a dry week over a wet one and the rain disappears. The analyst never sees a forged document, only a genuine, correctly signed series that happens to be false.

2.2 The three standard attacks

- **Step bias, 12 %.** From one day on, every value is raised or lowered by a fixed proportion. It is hard to see because 12 % is about the normal difference between a point station and a 50 km reference grid cell.
- **Gradual drift, up to 15 %.** The offset grows slowly until it reaches its final size. It looks like a sensor ageing, which is a very common real fault.
- **Replay, 7 samples.** A recorded stretch of genuine past data is played back as if it were new. Each value is real weather, so no single value looks wrong.

"12 % of what?" The exact meaning is not fixed. So every attack was run with all three readings and every result is reported separately for each:

- **12 % of the value:** $x \to 1.12 \cdot x$. This is the natural reading.
- **12 % of the range:** $x \to x + 0.12 \cdot (\text{99th percentile} - \text{1st percentile})$. The range is measured on the days before the attack.
- **12 % of the standard deviation:** $x \to x + 0.12 \cdot (\text{standard deviation})$. This is a much smaller attack. The standard deviation is measured before the attack.

Range and standard deviation are measured on the days before the attack, so the attacker's own values cannot change its size.

2.3 Why stations are easy to attack

- **Unattended and reachable.** A person and a few minutes are enough.
- **The person with access may be the one who benefits.** An insurance policy pays out when the station records too little rain or too much heat, and the landowner often hosts the station.
- **Cheap and constrained.** Little power, small messages, so security is the first thing left out.
- **Sparse.** Few stations cover a large area, so one false station can move a decision.
- **Trusted by default.** Satellite products are validated against station data, so the stations are rarely questioned.
- **Faults look the same.** Humidity sensors drift, radiation sensors get dirty, rain gauges clog. A slow drift is both the most common fault and one of the standard attacks.

2.4 Integrity is not truthfulness

Integrity means the reading was not changed after the station produced it. Cryptography delivers that. Truthfulness means the reading reflects the real weather. Only cross-checking against independent evidence delivers that. Track 2 lives in the gap between the two, so the design has one layer for each.

3 Threat model

3.1 The chain, and where an attacker can stand

Readings travel from the field to a decision. Each step has a different level of trust, and the attacker can be at four different places:

- **P1a, at the sensor.** The attacker tricks the physical measurement: shades the radiation sensor, waters or covers the gauge, warms a probe, moves the station. Signatures do not help, because the station honestly signs what it measured. Cost to the attacker: very low.
- **P1b, in the logger or firmware.** The attacker changes values before they are signed (one calibration factor gives a 12 % bias) and may steal the key. Signatures do not help, because the false data carries a valid signature. Cost: medium.
- **P2, on the link, gateway or platform.** The attacker reads, injects, modifies, replays, delays and drops messages. Signatures help against forgery, modification and replay. Dropping is detected, not prevented. Cost: low to medium.
- **P3, in the ingest, database or processing.** The attacker rewrites stored or forwarded values and deletes records. Signatures help if verification is end to end and the ledger is checked. Cost: medium (intrusion or insider).

Three attacker knowledge levels were used in the evaluation:

- **Black box:** knows what the station measures, not the defence.
- **Grey box:** knows the design, not the thresholds.
- **White box:** knows thresholds and models, and reads the same public NASA POWER data.

Assumed out of reach: the offline root key, breaking Ed25519 or SHA-256, and falsifying NASA POWER or Sentinel-2 at their source.

3.2 What is worth protecting

- **A measurement** needs integrity and truthfulness. If it fails, the indicator and the decision are wrong.
- **Where and who it came from** needs authenticity and provenance. If it fails, a reading claims a station or place it did not come from.
- **When it was measured** needs freshness and order. If it fails, last week's weather is presented as this week's.
- **The stream itself** needs availability and a way of noticing absence. If it fails, an event goes unrecorded and nobody notices.
- **Reference and context data** need integrity and a traceable origin. If it fails, the witnesses cannot be audited.
- **Processing history** needs auditability. If it fails, a decision cannot be traced back to its readings.

- **Station private keys** need confidentiality. If it fails, the attacker signs as the station.
- **Registry, replay state and ledger** need integrity. If it fails, verification can be subverted.
- **Calibration, baselines and models** need integrity. If it fails, the attacker teaches the detector that false is normal.

3.3 Catalogue of injection attacks

- **F1, step bias** (standard, 12 %). Positions P1, P2, P3. Evaluated, 288 runs.
- **F2, gradual drift** (standard, up to 15 %). Positions P1, P2, P3. Evaluated, 432 runs, plus a slower-drift test.
- **F3, replay** (standard, 7 samples). Positions P1b, P2, P3. Evaluated, 36 runs, plus a noisy-replay test.
- **F4, freeze** (last value repeated). Positions P1, P2. Caught by the replay matcher at lag 1; not tested separately.
- **F5, event masking or fabrication** (a rainy week erased, a heatwave inserted). Positions P1b, P2, P3. Planned for later work.
- **F6, noise or spike injection.** Positions P1, P2. Planned for later work.
- **F7, suppression** (readings dropped). Positions P1, P2, P3. Caught by the chain gap and the cadence watchdog (unit tests).
- **F8, delay or date shift.** Positions P2, P3. Caught by the timing check (unit tests).
- **F9, coordinated injection over several variables or stations.** Several positions. A replay at P1 copies all variables; otherwise planned for later work.
- **F10, provenance stripping** (signature removed). Positions P2, P3. Prevented: unsigned records are never accepted.
- **F11, station key compromise.** Position P1b. Same as P1; the detection results apply unchanged.

3.4 How a smart attacker might try to dodge

- **E1, slower and smaller drift** stretched over months. Tested in Section 7.5.
- **E2, stay inside range and rate-of-change limits.** The standard attacks already do; detection never relies on limits.
- **E3, change temperature, humidity and radiation together** so they stay consistent. Planned for later work.
- **E4, analogue replay:** replay the past week that best matches the current NASA POWER week. Planned for later work.
- **E5, noisy replay,** so the copy is not exact. Tested in Section 7.5.
- **E6, hide in the reference's own error** (rain, wind, humidity). Measured as the "stealth budget" in Section 7.5.
- **E7, poison the baseline** by being present while the system learns. A stated limit of file mode (Section 9).
- **E8, fault mimicry and alarm fatigue.** Limited by the false-alarm budget and graded responses; not evaluated.
- **E9, white box:** the largest change that stays under every threshold. Approximated by the step-size sweep in Section 7.5.

3.5 Which layer stops which attack

- **Step bias.** Layer 1 prevents it at P2 and P3. Layers 2–3 use CUSUM against the reference, and Mahalanobis distance for big steps. Outcome: prevented in transit; detected at the station within days if above the natural spread.
- **Gradual drift.** Layer 1 prevents it at P2 and P3. Layers 2–3 use CUSUM once the offset grows past the spread. Outcome: detected late, but bounded.
- **Replay.** Layer 1 prevents it at P2 and P3 (sequence number, chain). Layers 2–3 use the replay matcher (an exact copy of the station's own past), CUSUM and Mahalanobis distance. Outcome: prevented in transit; detected on day 1 at the station.
- **Freeze.** Layer 1 does nothing. The replay matcher catches it at lag 1. Outcome: detected.
- **Suppression.** Layer 1 detects it (chain gap, cadence watchdog). Layers 2–3 do nothing. Outcome: detected, not prevented.
- **Delay.** Layer 1 detects it (date against arrival, sequence). Layers 2–3 do nothing. Outcome: detected.
- **Stripping.** Layer 1 prevents it. Layers 2–3 do nothing. Outcome: prevented.
- **Key compromise.** Layer 1 is defeated, limited by revocation. Layers 2–3 apply unchanged. Outcome: detected by content.

Signatures alone close the telemetry and the pipeline completely, and do nothing at the station. Detection against the witness catches a lying station however the lie was made, but only as sharply as the reference represents that variable at that place. That is the argument for having both.

3.6 Residual risk, stated honestly

- A bias smaller than the reference's own error is undetectable. The defence limits how far an attacker can move the picture; it does not make that limit zero. Section 7.5 measures it per variable.
- Daily rain and wind at a point are poorly represented by a grid cell, so these variables are weakly protected.
- Attacks and faults look alike. The system detects that a stream is wrong, not why. The response is the same.
- Availability cannot be guaranteed. Suppression is detected and compensated, not prevented.
- The reference is not perfectly independent (a reanalysis may use public observations). This is to be checked per variable in later work.
- Out of scope: attacks on satellites or on NASA POWER or Copernicus themselves, confidentiality, denial of service against the trust service, and the correctness of the farm or insurance model downstream.

3.7 Is the threat real?

Yes, and similar attacks are documented. Coordinated false data can pass residual-based checks in power-grid state estimation [1], which is why we rely on an independent witness and not on the sensor's own consistency.

Stuxnet replayed recorded normal readings to operators while a process was driven out of bounds [2], which is the replay attack studied here. Sound and electromagnetic interference can inject signals into physical sensors [3, 4], which is the P1a attacker.

Quality control of automatic weather stations is a known operational problem [5], and the FAO-56 method behind the decision indicator [6] is sensitive to exactly the variables attacked

here.

4 Architecture and trust anchor

4.1 Six design rules

- **Rule 1. Sign at the source, verify at the service.** The station key never leaves the station and every record is checked before any analysis.
- **Rule 2. Look for disagreement, not extremes.** Real extreme weather moves the station and its witnesses together. An injection moves only the station.
- **Rule 3. Collect evidence over days.** A 12 % bias hides in any single day, but a disagreement that never averages out does not.
- **Rule 4. Set thresholds from a false-alarm budget** using clean training years only, so the cost for honest stations is fixed before any attack is seen.
- **Rule 5. Trust is a score, not a switch,** and every action comes with a reason an operator can check.
- **Rule 6. Nothing depends on a particular station.** The input is a series plus coordinates, because the station to be checked may be one the system has never seen.

4.2 The whole pipeline

A reading moves along this path: station, signed record, layer 1 check, comparison with NASA POWER, detectors, trust and response, decision. A single reading goes through these steps:

- **Step 1, station.** Measures the day and builds a record: sensor id, key id, sequence number n, observation time, hash of record $n-1$, values. Signs it with its Ed25519 key.
- **Step 2, link.** Carries 139 bytes (for 5 variables). An attacker here has no key.
- **Step 3, layer 1.** Checks the certificate (enrolled, not revoked, valid at that date, position bound), the signature, the sequence number, the time against arrival and cadence, and the previous hash. Then appends to the hash-chained ledger.
- **Step 4, preparation.** Joins the day with NASA POWER for the enrolled position, applies the frozen per-season calibration, computes the standardized residual per variable.
- **Step 5, detection.** Updates the CUSUM sums, replay distances, Mahalanobis distance, trend statistic and detector M features.
- **Step 6, fusion and trust.** Combines the day's evidence into a compromise probability p, updates the trust score, maps it to a band.
- **Step 7, response.** Trusted: full weight. De-weighted: affected variables blended toward the calibrated reference, with a verification request if trust keeps falling. Quarantined: affected variables replaced, alert with reason and estimated onset.
- **Step 8, product.** Computes ET0 and the 30-day water balance after the response, and issues the decision as TRUSTED or DEGRADED with its provenance record.

4.3 Layer 1: the trust anchor

- **Root of trust.** An offline Ed25519 root key held by the network operator. It only signs certificates.
- **One key per station,** created on the device. The private key never leaves it. In the prototype it sits behind a SecureElement interface that only exposes `sign()` and

`public_key()`, which is exactly what real secure-element hardware would offer. Breaking into one station does not let an attacker speak for another, and every false reading is traceable to one device.

- **Enrolment binds the position.** The root signs a certificate linking the station's public key to its id, its position and a validity window. The reference data is looked up by position, so a station that could claim another place would be checked against the wrong witness.
- **Rotation.** Every record carries a 2-byte key id. A new certificate with a new key id replaces the old one at a given date.
- **Revocation.** A compromised station stays in the registry (its past remains on record) but its later records are rejected.

4.4 What a record looks like

Field	Size	Purpose
version	1 B	format evolution
type	1 B	reading, status, heartbeat
sensor_id	4 B	which station
key_id	2 B	which station key (rotation)
seq	4 B	strictly increasing: ordering and replay protection
t_obs	8 B	observation date at the station
prev_hash	32 B	SHA-256 of the previous record: detects gaps, reordering, history rewriting
payload_len	2 B	length of the payload
payload	variable	the day's values as fixed-point numbers (value × 1000, 32 bits) plus a status byte
signature	64 B	Ed25519 over every field above

Table 1: Fields of a signed record.

The fixed part is 118 bytes (checked by a unit test). With five variables the payload is 21 bytes and the whole record 139 bytes. At one record per station per day this is negligible on a cellular link. On a low-power radio limited to a few tens of bytes, the documented option is a compact mode (truncated chain hash, one signature per batch of days).

4.5 How the service checks a record

- **Check 1.** Parse the record. If malformed: reject (malformed).
- **Check 2.** Unknown or revoked station, certificate not valid at that date, or key id mismatch: reject (unknown_station).
- **Check 3.** Invalid Ed25519 signature: reject (forged).
- **Check 4.** Sequence number not above the last accepted one: reject (replay).
- **Check 5.** Observation date later than arrival + 1 hour: reject (stale). Arrival more than 2 days after the observation (cadence 1 day + margin 1 day): flag as delayed.
- **Check 6.** Previous hash does not match the last accepted record: accept, but raise a GAP.
- **Check 7.** Save last_seq, last_hash and last_arrival, pass the reading on: accept.

Per-station state is just two values (last_seq, last_hash) plus the last arrival time. A station with no valid record in the expected interval plus a margin is reported silent: its weight is removed and product confidence drops. Silence is never read as an uneventful day.

Tamper-evident ledger. Accepted records are appended to a hash chain: entry i stores SHA-256(previous entry hash $\|$ i $\|$ record). Rewriting any stored record changes its hash and breaks every later link, which is how an attacker inside the pipeline (P3) is caught.

Provenance into the product. Every product carries a provenance record:

- the inputs used (sensor, sequence range, chain hash, weight);
- the inputs excluded (sensor, reason, estimated onset);
- the NASA POWER request (parameters, point, dates, SHA-256 of the cached response);
- the processing steps and parameters;
- the confidence (TRUSTED or DEGRADED, with the reason).

4.6 Layer 1 tested: the same attacks at three positions

Every day of 2021–2024 at the Florida station was signed with a real Ed25519 key, passed through the attacker, checked by the real verifier and stored in the real ledger.

Attack	Pos.	Days	Rejected	Ledger intact?	Reach detection
Step 12 % temperature	P1	60	0	yes	60
Step 12 % temperature	P2	60	60 (forged)	yes	0
Step 12 % temperature	P3	60	0	no, broken at 2024-04-15	60
Replay 7 days	P1	7	0	yes	7
Replay 7 days	P2	7	7 (replay)	yes	0
Replay 7 days	P3	7	0	no, broken at 2024-04-15	7

Table 2: Layer 1 against the same attacks at three positions.

On the link (P2) every tampered record fails its signature and every replayed record fails the sequence check, so nothing attacked reaches detection. In the pipeline (P3) the rewritten values do reach the store, but the ledger breaks at the first rewritten day and points to it. At the station (P1) layer 1 accepts everything. That case belongs to detection, and it is the one evaluated in Sections 6 and 7.

4.7 What it costs

- Record size, fixed part (header + signature): 118 bytes.
- Payload, 5 variables: 21 bytes.
- Record size, total: 139 bytes (overhead 5.62 times the payload).
- Sign one record (station side): 62.4 µs.
- Verify one record (signature, certificate, sequence, chain): 150.5 µs.
- Append one record to the ledger: 2.4 µs.
- Whole defence on one station, 4 years of daily data: 20.96 s.
- Whole defence per daily reading: 14.35 ms.
- Peak Python memory of that run: 29.5 MiB.
- Detector S fitted state per station: 1,444 bytes.
- Replay-matcher history per station (730 days × 4 variables): 23,521 bytes.
- Detector M saved model (3 sites): 4.18 MiB.
- Fusion model and trust bands: 2,089 bytes.

Measured on an 11th Gen Intel Core i5-1135G7 at 2.40 GHz, 19.7 GB RAM, Windows 11, Python 3.12.10. Software only, standard primitives, no special hardware.

5 Data and signal processing

5.1 The data

Every measurement is real, from public datasets, and attacks are injected into real records, so the labels are exact to the day. All five variables are used at all three sites from 1 January 2021 to 31 December 2024.

- **AZMET** (University of Arizona), station az09. Arizona, USA, semi-arid climate, daily data. Latitude 32.480, longitude −109.930.
- **FAWN** (Florida Automated Weather Network), Okahumpka. Florida, USA, humid subtropical climate, 15-minute data averaged to daily. Latitude 28.682, longitude −81.886.
- **Kansas Mesonet,** Manhattan. Kansas, USA, continental climate, daily data. Latitude 39.209, longitude −96.592.
- **NASA POWER Daily API** (the reference), community AG. The same three points. Reanalysis plus satellite data on a grid of about 0.5°, daily.

- **Training period:** 2021–2023 (calibration, thresholds, models; clean data only).
- **Test period:** 2024. Every reported number is measured on it.
- **Sentinel-2 L2A** is part of the design but its indicators are not computed yet. This is a stated limitation and the first planned next step.

5.2 Cleaning

Every source is converted to one table with one row per calendar day. Missing values stay as missing (never 0, never −999) and nothing is interpolated.

- Fill codes and impossible values become missing, and every removal is counted. Valid ranges: temperature −40 to 50 °C, humidity 1–100 %, wind 0–40 m/s, rain 0–300 mm/day, solar 0–40 MJ/m^2/day.
- FAWN's 15-minute data is averaged per day (temperature, humidity, wind, solar) or summed (rain). A day needs at least 80 % of its readings, variable by variable.
- Units are converted (mph → m/s, inch → mm, W/m^2 × 0.0864 → MJ/m^2/day).
- Each NASA POWER file must sit at the coordinates of a known site. Responses are cached with SHA-256 manifests.

5.3 A point against a grid cell: calibration

A station measures a point and NASA POWER describes a cell of about 50 km, so they differ. Part of the difference is systematic (altitude, exposure, local climate) and part is random. The calibration learns the systematic part on the training years, for each variable and each season (DJF, MAM, JJA, SON), using robust statistics (medians):

- **gap** = station − power (temperature, humidity, solar);
- **gap** = $\log\left((\text{station} + 0.1)/(\text{power} + 0.1)\right)$ (wind, rain: errors are multiplicative);
- **offset** = median of the gap on training days of that season;
- **spread** = $1.4826 \times \text{median}(|\text{gap} - \text{offset}|)$, a robust standard deviation;

– **residual** = (gap − offset)/spread, measured in standard deviations.

A residual of 0 means "the station agrees with the reference as usual". A residual of +3 means "three normal spreads above what we expect".

A sharper witness for humidity. Humidity is very local. NASA POWER's humidity follows the station only loosely (correlation 0.76–0.90, against 0.99 for temperature), so a 12 % humidity attack was only 1–1.7 times the normal day-to-day disagreement, against 2.6–5.7 times for temperature.

For humidity the witness is therefore a ridge regression of station humidity on same-day POWER values (humidity, temperature, dew point, log rain, wind, solar and the season), fitted on clean training days only (15 features, $\alpha = 10$). Only POWER values go into it, so an attack on the station cannot move its own witness, and a temperature attack does not leak into the humidity check.

Tested out of sample (fit 2021–2022, measured on 2023), it raised the ratio of a month-long 12 % humidity shift to the natural month-long wandering from 2.0 to 3.1 (AZMET), 2.8 to 3.5 (FAWN) and 2.3 to 4.1 (Kansas). A dew-point comparison was tried and rejected: worse at AZMET and it mixes temperature attacks into humidity.

Rain has a special rule. A day when both the station and POWER are dry (0.1 mm or less) carries no information, so its residual is missing, not zero. This applies to 40–70 % of days depending on the site. The "calibrated reference" used for substitution in quarantine is the opposite operation: POWER plus the offset (or the inverse log ratio).

Temporal residuals. A second family compares each station with its own expected value for the season (harmonic regression, order 2) and its recent history (AR(1)). These are computed and validated, but the detectors use the reference residuals, because only an independent witness separates a lie from real weather.

5.4 Validation gate: do the residuals behave on clean data?

On the clean test year 2024 (never seen by the calibration) the residuals are centred near zero and have a similar width in all three climates and for every variable. For each site, variable and season the preparation stage checks centring (median within 0.25 standard deviations), width compared with training, inflation during extreme weather, and autocorrelation.

Site	PASS	WARN	FAIL
AZMET	44	16	5
FAWN	27	18	20
Kansas	44	16	5

Table 3: Validation checks on clean 2024 data.

The remaining FAILs are level shifts (the mean of a season sits off zero) and autocorrelation, not inflated widths. That observation turned out to be the central fact of the whole design.

The central finding. With no attack at all, the station-minus-grid disagreement wanders by up to about ±1 standard deviation for one to three months at a time. At FAWN in 2024 the clean temperature residual averaged +1.45 standard deviations in March, +1.43 in October and

−0.83 in July. A 12 % bias is 1 to 4 standard deviations. So no fixed threshold can separate a 12 % attack from natural wandering at every site.

A detector has to accumulate evidence over days, and its threshold has to be an openly stated trade-off between false alarms and detection, set from data. A first design that removed a slow baseline (180-day median, 60-day gap) was built and measured. It did not lower false alarms overall (FAWN 4 → 2, Kansas 3 → 5 per year), so it is kept off.

How big is a 12 % attack in residual units? This is the 12 % of the mean value divided by the fitted spread, expressed in standard deviations, and it predicts detectability (Section 7.3).

12 % of the mean value, in standard deviations	AZMET	FAWN	Kansas
temperature	1.23	3.19	1.10
humidity	0.90	3.03	1.15
wind	0.55	0.64	0.53
solar	3.00	1.34	1.57
rain	0.08	0.06	0.07

Table 4: Size of a 12 % attack in residual units.

5.5 Replay check and change-point dating

A replayed week is invisible in a single day's values, since they are genuine weather, and does not repeat in the residuals either, since the reference has moved on. It leaves one trace: an exact copy of the station's own earlier values.

For every 7-day window and every lag from 1 to 730 days, the matcher computes the average distance (over 7 days and 4 variables, in training standard deviations) between the window and the same window one lag earlier. If the smallest distance is below 0.05, it flags a replay.

On clean training data the smallest natural distance was 0.10 (AZMET), 0.20 (FAWN) and 0.19 (Kansas), which is two to four times the tolerance. The matcher keeps the last 730 training days so a replay early in the year still finds its source, and lag 1 also catches a frozen sensor (F4).

Dating an attack. Each CUSUM alarm is dated back to the day after the alarming sum was last at zero (the classic change-point estimate). Alarms less than 14 days apart form one episode with an estimated onset, a direction, the variable, and a type: "drift" if a Mann-Kendall trend test is significant at 5 %, otherwise "step". The median onset error is 2–6 days for steps.

5.6 Specified but not built yet

Physical-consistency residuals between variables (beyond the humidity witness) and residuals against Sentinel-2 indicators (cloud, vegetation, moisture around the site) are specified but not implemented yet. They are the two extra witnesses that would address the small-attack limit.

6 Detection

Two detectors work side by side. S is statistical and needs no learning beyond simple calibration. M is a learned one-class model. Their outputs are fused into one probability.

6.1 Detector S, the statistical baseline

Whitening. The residuals are autocorrelated (lag-1 coefficient 0.1–0.7): today looks like yesterday. An AR(1) model fitted per variable on the training years predicts each day from the previous one, and the prediction error divided by its training standard deviation is close to independent noise of variance 1, which is what CUSUM theory expects. A lasting offset δ survives whitening as $(1-\varphi)\,\delta/\sigma_e$, where φ is the day-to-day carry-over and σ_e the standard deviation of the prediction error.

CUSUM with restart (Page's test), two-sided, one pair of sums per variable, slack $k = 0.5$ standard deviations:

$$U(t) = \max\big(0,\, U(t-1) + w(t) - k\big), \qquad D(t) = \min\big(0,\, D(t-1) + w(t) + k\big),$$

$$S(t) = \max_{\text{variables}} \max(U, -D), \qquad \text{alarm if } S(t) > h, \text{ then every sum restarts at 0.}$$

Here $w(t)$ is the whitened residual of the day. In words: noise inside $\pm k$ cancels out, while an offset bigger than k builds up day after day. A lasting attack therefore produces repeated alarms. A missing day holds the sums (never filled).

The threshold h is not guessed. It is the lowest value that gives at most one alarm per year on the clean training years, found by bisection. It adapts to each site: $h = 8.9$ (AZMET), 14.9 (FAWN, the site that wanders most) and 8.6 (Kansas). This matters because the station to be checked may be new.

Mahalanobis distance catches one unusual day across variables. It uses a robust centre and scale (median, MAD) and a covariance from the training residuals with 10 % shrinkage toward the diagonal. The squared distance is compared with the χ^2 quantile at 0.999 with 4 degrees of freedom (18.47).

Replay matcher. See Section 5.5.

Trend. Rolling Mann-Kendall z over 150 days on the whitened residuals. The threshold is the larger of the Bonferroni-corrected value ($|z|$ above 3.66 for $\alpha = 0.001$ over 4 variables) and the value allowing one alarm per year on training. It still flags 9–20 % of clean days at two sites, because a 150-day window stays above the threshold for weeks. So it is reported but left out of the combined flag.

Combined flag. Detector S raises an alarm if CUSUM or the replay matcher or Mahalanobis fires. Rain is not monitored, since its residual is undefined on dry days.

6.2 Detector M, the learned detector

A one-class model trained on clean windows only. Its features for each day are all causal (nothing from the future):

- for each residual: its level, 1-day change, 7- and 30-day means, 7-day standard deviation and slope, and 30-day sum;
- the 7-day correlation of station and POWER day-to-day changes;
- 7-day replay distances at offsets 7, 28 and 364 days;
- simple quality flags and missing-value indicators.

2024 is never seen by M before the evaluation.

- **Model choice.** Six PyOD models (Isolation Forest, ECOD, COPOD, HBOS, KNN, LOF) were fitted on clean 2021–2022 windows and compared on 129 attacks injected into 2023. LOF came first with a mean ROC-AUC of 0.65 against 0.49–0.64 for the others.
- **Thresholds, leaving one year out.** LOF is fitted on two training years and scores the third, three times, so every clean training day gets an out-of-sample score. The threshold is the 99th percentile of those scores. A first version set thresholds from 2023 alone and flagged 6 % of FAWN's 2021 days, because residuals wander from year to year and a percentile from about 90 days rests on one or two extreme values.
- **Granularity.** Global, per site or per site and season, whichever keeps the false-alarm rate nearest 1 % on a held-out training year. Mean / worst rate: global 2.9 % / 19 %, per site 3.2 % / 22 %, per season 6.0 % / 28 %. Global is used. The worst case shows how much one year can differ from the others.
- **Final model.** LOF fitted on every clean 2021–2023 window.

Caveat on M. M's replay features look back 7, 28 and 364 days, which include the offsets used in the replay scenarios, so its replay result is somewhat favourable. M exists only for the three development sites (its calibration is per site). Any other station uses the S-only fusion.

6.3 Fusion: one calibrated compromise probability

The daily outputs of S and M become nine features:

- `cusum_ratio`: the CUSUM statistic divided by its threshold;
- `cusum_recent`: alarms in the last 30 days;
- `maha_log` and `replay_log`: the Mahalanobis and replay evidence;
- `trend_ratio`: the trend evidence;
- `gap7` and `gap30`: the largest 7- and 30-day average residual, relative to the station's own normal level (a drift is invisible day by day but clear as a month-long average);
- `m_flag` and `m_recent`: the M alarm today and recently.

A logistic regression turns them into a probability p. It is fitted on labelled runs (attacked days = 1, all others = 0). Leave one site out: the model used at a site is fitted on the two other sites only, exactly as a new site will be handled.

Fitted weight	Held out: AZMET	Held out: FAWN	Held out: Kansas
intercept	−2.87	−3.64	−2.68
`cusum_ratio`	−0.05	0.28	0.69
`cusum_recent`	0.17	−0.03	0.45
`maha_log`	0.90	0.55	0.73
`replay_log`	0.23	0.01	0.35
`trend_ratio`	−0.57	−0.40	−1.40
`gap7`	1.34	1.05	1.67
`gap30`	1.51	2.07	1.54
`m_flag`	0.73	0.29	0.20
`m_recent`	−0.14	−0.20	−0.15

Table 5: Fusion weights fitted with one site held out.

The three fits agree and are easy to read: the week- and month-long gaps weigh most, and an unusual day and an M alarm also raise the probability. The trend feature gets a negative

weight, because once the gaps are known a high trend score mostly marks natural wandering.

Is the probability honest? On the held-out sites (277,794 days), days scored 0.3–0.4 were attacked 30 % of the time, and days scored 0.9–1.0 were attacked 95 % of the time. The Brier score is 0.131 against 0.162 for always predicting the base rate. The probability is relative to the share of attacked days in the evaluation runs (20 %). A deployment with rarer attacks shifts it by Bayes' rule on the odds, which the code provides.

6.4 What is fitted on which data

- **Calibration, AR models, CUSUM / trend / Mahalanobis thresholds, replay history:** clean 2021–2023 of each site. Never sees 2024 or any attack.
- **Detector M model type:** clean 2021–2022, compared on attacks injected in 2023. LOF chosen.
- **Detector M thresholds and final model:** clean 2021–2023, leave one year out; final fit on all three years. Never sees 2024.
- **Humidity witness regression:** clean 2021–2023 of each site, POWER inputs only. Part of the calibration.
- **Fusion and trust bands (evaluation):** attacked and clean runs of the two other sites. Leave one site out.
- **Fusion and trust bands (deployment):** all three sites, frozen. S-only variant for stations without M.
- **Every reported number:** 2024, held-out site where relevant. Attacks only ever in 2024 (checked by the runner).

6.5 Day-by-day accuracy and which evidence matters

These scores count every attacked day, including the first days of an attack and the many attacks too small to be seen. They therefore understate episode-level detection, which is reported in Section 7.

Fusion variant	ROC-AUC	PR-AUC	TPR 1 %	TPR 0.1 %	Step	Drift	Replay
full fusion (S + M)	0.728	0.530	0.188	0.085	0.743	0.721	0.952
S only	0.721	0.521	0.184	0.070	0.747	0.708	0.956
M only	0.534	0.264	0.070	0.012	0.547	0.526	0.881
CUSUM only	0.704	0.458	0.141	0.031	0.698	0.705	0.864
without CUSUM	0.733	0.537	0.190	0.089	0.745	0.727	0.958
without Mahalanobis	0.727	0.527	0.188	0.079	0.741	0.719	0.926
without replay matcher	0.734	0.531	0.181	0.086	0.745	0.728	0.920
without trend	0.728	0.538	0.207	0.086	0.737	0.722	0.963
without rolling gaps	0.699	0.476	0.163	0.074	0.696	0.699	0.969

Table 6: Per-day scores on the held-out site, with one group of evidence removed at a time. TPR 1 % and TPR 0.1 % are the share of attacked days caught at 1 % and 0.1 % false positives. Step, Drift and Replay are the ROC-AUC for each attack.

- **Rolling gaps matter most** (AUC 0.728 → 0.699 without them). Accumulating disagreement over days is the core of detection. It enters twice, through CUSUM and through the month-long gap features, so removing only one of them costs little (without CUSUM: 0.733).

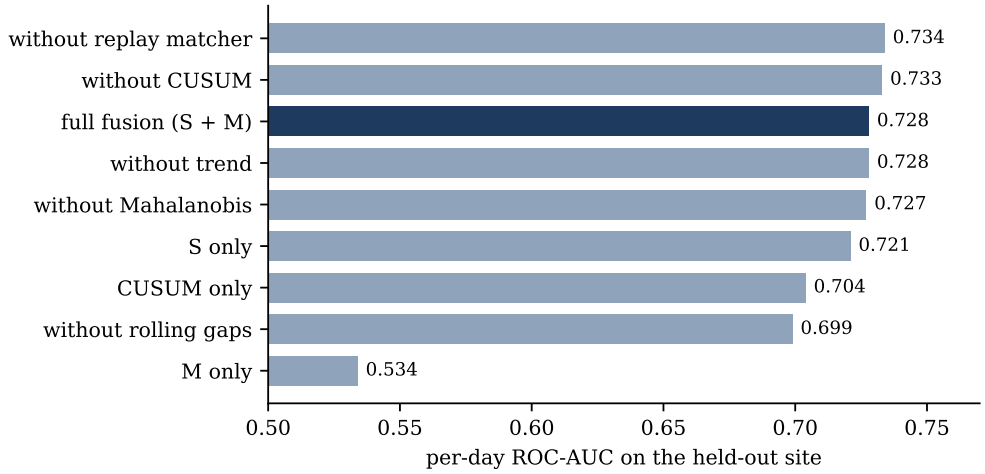


Figure 1: Per-day ROC-AUC on the held-out site, with one group of evidence removed at a time.

- **CUSUM stays** for what the AUC does not measure: it dates the onset, names the variable and the direction, and drives the alert.
- **M adds a little on top of S** (0.721 → 0.728; TPR at 1 % FPR 18 → 19 %). Alone, tuned without 2024, it reaches only 0.534.
- **Removing the replay matcher** does not lower the AUC, even for replays, because a replayed week also disagrees with NASA POWER. Its value is elsewhere: it names the attack, dates it to the day, and has zero false alarms.
- **The ceiling of about 0.73** comes from the many attacks smaller than the natural disagreement (Section 7.3), not from the fusion.
- **Sentinel-2** cannot be ablated because it is not computed yet. That is a future ablation.

7 Testing against attacks, and the response

7.1 The standard scenario

File mode, attacker at the station (P1): the series arrives with the injection already inside. Per site, calibration and detectors are fitted once on clean 2021–2023, then every attack is injected into 2024 and scored with the same fitted models. This is exact, because attacks never touch the training years.

- **12 % step, 60 days.** Variables: temperature, humidity, wind, solar. Signs: up and down. Meaning of %: value, range, standard deviation. Four onsets (one per season). 96 runs per site.
- **Drift to 15 %, held 30 days.** The same 4 variables, up and down, with the three meanings. Ramp of 30, 60 or 90 days, with 2 onsets. 144 runs per site.
- **7-day replay.** Whole record. Source 7, 28 or 364 days back, with 4 onsets. 12 runs per site.

That is 252 runs per site and **756 attacked runs** in total.

Fair detection. A detector with a few false alarms a year will sometimes alarm inside an attack window by chance. So every attacked run is compared with the clean run of the same site: an alarm only counts for the attack if the clean run has no alarm that day. An episode is detected if such an alarm falls between its first and last attacked day. Rates come with 95 %

Wilson confidence intervals.

7.2 Results

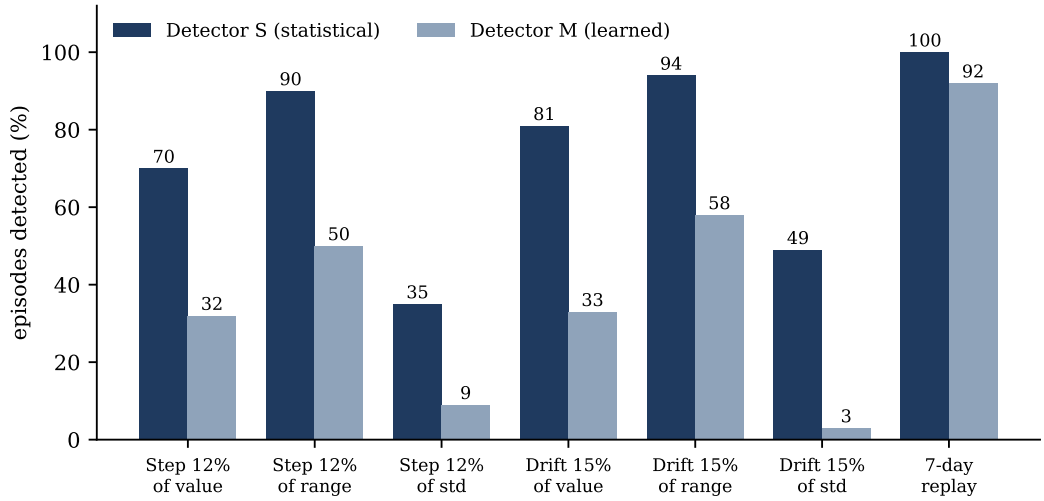


Figure 2: Share of attack episodes detected, 756 runs over 3 sites in 2024.

Attack	Detected (95 % CI)	Latency	Onset error
12 % step, of value	67/96 = 70 % (60–78)	12 d	3 d
12 % step, of range	86/96 = 90 % (82–94)	6 d	2 d
12 % step, of std	34/96 = 35 % (27–45)	28 d	6 d
Drift 15 %, of value	117/144 = 81 % (74–87)	40 d	27 d
Drift 15 %, of range	136/144 = 94 % (89–97)	30 d	24 d
Drift 15 %, of std	71/144 = 49 % (41–57)	48 d	36 d
7-sample replay	36/36 = 100 % (90–100)	0 d	1 d

Table 7: Detection of attack episodes by detector S. “Std” stands for the standard deviation.

Attack	M detected	M latency	S or M
12 % step, of value	31/96 = 32 %	19 d	72/96 = 75 %
12 % step, of range	48/96 = 50 %	16 d	87/96 = 91 %
12 % step, of std	9/96 = 9 %	5 d	37/96 = 39 %
Drift 15 %, of value	48/144 = 33 %	54 d	120/144 = 83 %
Drift 15 %, of range	84/144 = 58 %	54 d	137/144 = 95 %
Drift 15 %, of std	5/144 = 3 %	102 d	73/144 = 51 %
7-sample replay	33/36 = 92 %	0 d	36/36 = 100 %

Table 8: Detection of attack episodes by detector M, and by S or M together.

Latency is the number of days from the attack start to the first alarm (median, detected episodes only). 0 d means flagged on the first attacked day.

7.3 What decides detection

One thing explains almost everything: how big the attack is compared with the natural station-to-reference disagreement.

Variable	step, value	step, range	step, std	drift, value	drift, range	drift, std
temperature	83 %	100 %	46 %	97 %	100 %	67 %
solar	88 %	96 %	54 %	97 %	100 %	69 %
humidity	71 %	75 %	21 %	78 %	81 %	25 %
wind	38 %	88 %	21 %	53 %	97 %	36 %

Table 9: Share of steps and drifts detected by detector S, by variable.

- Above about 1.5 standard deviations of the station-minus-reference difference, steps are almost always caught. Below about 0.7 they hide in the natural wandering.
- Attacks of 1.6–3.0 standard deviations (solar of range, solar of value, temperature of range, temperature of value) are almost always caught. Those of 0.3–0.9 (humidity, solar, temperature and wind of the standard deviation, and wind of value) mostly hide.
- With the regression witness, humidity attacks of value or range are now 1.2–1.3 standard deviations.
- The sign of the attack (up 62 %, down 67 % of steps) and the onset season (60–74 %) make little difference.

7.4 False alarms, including real extreme weather

Alarm events per year, clean 2024	AZMET	FAWN	Kansas
S (combined)	3.0	5.0	6.0
S: cusum	1.0	4.0	3.0
S: mahalanobis	2.0	1.0	3.0
S: replay	0.0	0.0	0.0
S: trend (not in the flag)	0.0	7.0	4.0
M	3.0	2.0	0.0
S or M	6.0	7.0	6.0

Table 10: False alarm events per year on clean 2024 data.

Detector S raises 4.7 alarm events per station-year (1.3 % of days) and the replay matcher none. The CUSUM budget was one per year on 2021–2023, and 2024 gave 1–4: the test year wanders more than the training years. This gap is reported, not hidden by retuning on 2024.

Extreme weather. A naive anomaly detector fails on genuine extremes, so we checked (temperature beyond the 5 % tails of its monthly climate, or rain above the 95th percentile of wet days, 34–44 days per site). Extreme days are flagged more often than ordinary ones at some sites: Kansas detector S 11.8 % against 0.6 %, and Kansas not trusted on 21 % of extreme days against 5 %.

The numbers are small but the direction is clear: the reference represents extremes less well than ordinary days. Ours fails this test mildly. Season- and regime-dependent spreads are a planned improvement.

7.5 Attackers who adapt

Stealth budget: how big a step passes unnoticed? (E6, approximating E9) Steps of 2 to 30 % were injected into each variable at all three sites. This is the smallest step detected in at least half of the runs:

– **Temperature:** 8 % of the value, 5 % of the range.
– **Solar:** 5 % of the value, 5 % of the range.
– **Humidity:** 16 % of the value, 12 % of the range.
– **Wind:** 8 % of the value, 8 % of the range.

For temperature and solar a 5–8 % step is already caught in half the runs and 8–12 % in all of them, so the standard 12 % is well outside the attacker's budget. For humidity the threshold is 12–16 %, so the standard 12 % humidity step is inside the budget (or right at its edge). Wind is caught in about half of the runs from 8 % upward but never reliably when read relative to the value. This is the numeric version of the residual risk in Section 3.6. (Detections at 2 % come from one site in each case and may be partly an alarm-shift effect.)

Slower drift (E1). Drifts to +15 % of the range were stretched from 30 to 270 days:

Days to reach +15 %	30	60	90	180	270
Detected (all variables)	58 %	92 %	83 %	92 %	92 %
Median latency	17 d	32 d	46 d	64 d	72 d

Table 11: Detection of slower drifts.

A slower drift is not detected less, it is detected more often, because it stays in the data longer and the CUSUM compares with an independent witness, not with the station's own past, so there is nothing to wait out. It is detected later, though: the median delay grows from 17 days to 72 days. That delay is what the attacker buys by going slowly, and it is the number to compare with the time a drift needs to change a decision. Humidity stays the weak variable at every ramp length.

Noisy replay (E5). A 7-day replay copied from 28 days back, with Gaussian noise added to every value (3 sites, 4 onsets). The noise level is given in standard deviations of each variable.

Noise	0	0.01	0.03	0.05	0.1	0.2	0.4
Replay matcher detects	100 %	100 %	100 %	100 %	0 %	0 %	0 %
Detector S as a whole detects	100 %	100 %	100 %	100 %	100 %	100 %	100 %

Table 12: Noisy replay against the replay matcher and against detector S as a whole.

Noise of 0.1 standard deviations defeats the exact-copy matcher, as expected from its 0.05 tolerance. But detector S as a whole still catches every noisy replay, through CUSUM and Mahalanobis distance: the replayed week no longer follows NASA POWER's weather for those days, and noise cannot hide that. The trick works against one detector and fails against the defence, which is the point of layering independent evidence. The analogue replay (E4) is the harder variant and is planned.

Evading layer 1. Stripping the signature is rejected, a forged or changed record fails its signature, a replayed or reordered one fails the sequence check, and a dropped one raises a chain gap and the cadence watchdog (all unit-tested). The only way around layer 1 is a stolen station key (F11), which turns the attacker into P1, the case evaluated above.

7.6 The response policy

Trust dynamics. Each day: $\text{risk}(t) = \text{risk}(t-1) + a \cdot (p(t) - \text{risk}(t-1))$ and trust = 1− risk. The rate a is 0.5 when p is above the current risk (trust falls fast) and 0.05 otherwise (half of the lost trust comes back in about 14 quiet days). One day at $p = 0.9$ takes trust from 1 to 0.55, and an attacker cannot reset trust by pausing for a day.

Bands, set on data and frozen. De-weighting starts below the trust level that clean days go under only 5 % of the time. Quarantine starts below the level they go under only 0.5 % of the time (on the fitting sites). There is hysteresis: release only above the de-weighting limit.

- **De-weighting** (trust in the de-weighted band). Value used = $w \cdot$ station + $(1 - w) \cdot$ calibrated reference, with w falling linearly from 1 to 0 across the band. Only for implicated variables (14-day mean residual beyond 1 standard deviation); the whole record for a detected replay.
- **Verification request** (de-weighted and trust still falling, at most every 14 days). Ask for independent evidence: next clear Sentinel-2 view, a neighbouring station, a technician visit.
- **Quarantine** (trust in the quarantined band). $w = 0$: implicated variables are replaced by the calibrated reference.
- **Alert on entering quarantine.** The operator is notified with the reason (the evidence that drove the probability; a detected replay is named explicitly), the variable, direction, type and estimated onset.

Replacing only the implicated variables matters. A first version replaced the whole record, rain included, and made decisions worse (wrong-decision days 14 % → 33 % on a test set), because substituting a grid cell's rain for a point gauge changes the water balance by itself.

Attack	De-weighted (95 % CI)	Median days	Quarantined + alert	Median days	Mean weight
drift, of range	126/144 = 88 % (81–92)	34	108/144 = 75 % (67–81)	40	0.56
drift, of std	57/144 = 40 % (32–48)	50	21/144 = 15 % (10–21)	47	0.91
drift, of value	106/144 = 74 % (66–80)	38	82/144 = 57 % (49–65)	40	0.68
replay	29/36 = 81 % (65–90)	0	30/36 = 83 % (68–92)	1	0.32
step, of range	78/96 = 81 % (72–88)	8	65/96 = 68 % (58–76)	13	0.46
step, of std	22/96 = 23 % (16–32)	31	12/96 = 12 % (7–21)	31	0.90
step, of value	63/96 = 66 % (56–74)	14	49/96 = 51 % (41–61)	19	0.61

Table 13: Responses during attacks. The last column is the mean weight given to the station during the attack.

Cost on honest stations (clean 2024 of each held-out site):

- **AZMET:** 3 % of days de-weighted, 0 quarantine days per year, 0 quarantines per year, 1 verification request per year, 0.0 % of decision days changed.
- **FAWN:** 0 % de-weighted, 0 quarantine days per year, 0 quarantines per year, 0 verification requests per year, 0.0 % of decision days changed.
- **Kansas:** 9 % de-weighted, 29 quarantine days per year, 1 quarantine per year, 2 verification requests per year, 1.1 % of decision days changed.

The quarantine level of a held-out site is the 0.5 % quantile of the clean trust of the two other sites, so it rests on about four days of clean data and moves with a single clean episode (AZMET 0.31, FAWN 0.40, Kansas 0.34). The only false-quarantine cost is one long natural disagreement at Kansas (29 days). The bands are the least stable part of the system, and getting more clean station-years to set them is the first planned improvement.

7.7 Farm decisions with and without the defence

Every series is turned into the decision it would drive. ET0 is computed by FAO-56 Penman-Monteith from temperature, humidity, wind and solar. The 30-day water balance is $B = \sum(\text{rain} - \text{ET0})$. Then a level per site and month, from the clean training years:

- **normal:** B at or above the 30th percentile;
- **dry (irrigation advised):** below the 30th percentile;
- **severe drought (the insurance trigger):** below the 10th percentile.

Three decision series are compared over the attack and the 30 days after it: from the true data, from the attacked data with no defence (C0), and after the responses (C4, full system). At P1, provenance alone (C2) changes nothing, so C2 = C0.

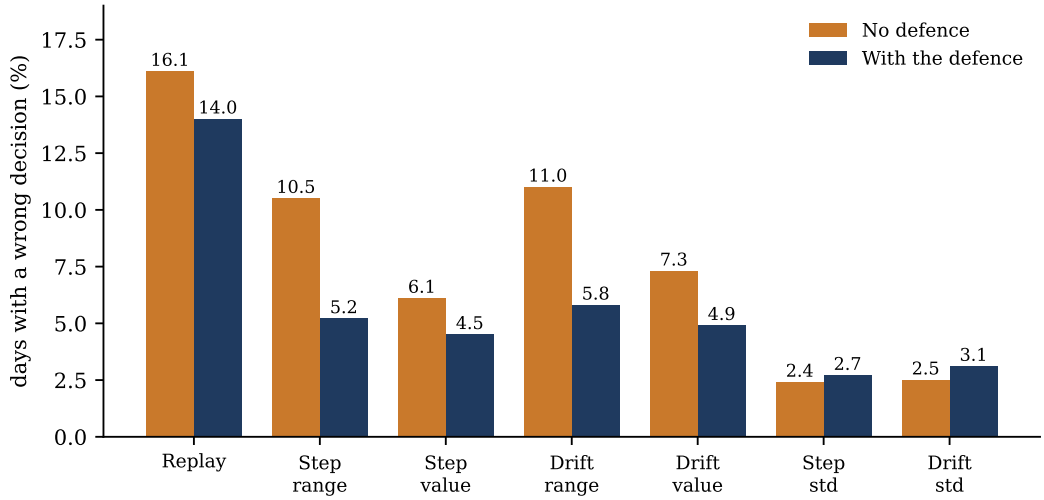


Figure 3: Share of days on which the farm decision is wrong because of the attack.

Attack	Wrong days, no defence	With the defence	Change
drift, of range	11.0 %	5.8 %	−47 %
drift, of std	2.5 %	3.1 %	+23 %
drift, of value	7.3 %	4.9 %	−33 %
replay	16.1 %	14.0 %	−13 %
step, of range	10.5 %	5.2 %	−50 %
step, of std	2.4 %	2.7 %	+14 %
step, of value	6.1 %	4.5 %	−26 %

Table 14: Share of days with a wrong farm decision, with and without the defence.

Overall the defence cuts wrong-decision days from 7.1 % to 4.9 % (−32 %). The more important effect is that **wrong decisions stop being silent**. Without the defence every wrong decision is issued as if sound. With it, a decision is TRUSTED only if the station was trusted over its whole 30-day window, otherwise it is DEGRADED (pending verification).

- 1 of the 36 replays still gave a wrong decision issued as trusted (21 replays changed the decision without defence). A replay is detected on day 1, but a station whose trust was already high can take a few days to fall below the de-weighting band.
- For the smallest attacks ("of std") the defence slightly *increases* wrong days (on average over steps and drifts, 2.5 → 2.9 %, see Table 14). They are rarely detected, and partial

Attack	Attacks that changed the decision	Of those, wrong and issued as TRUSTED
drift, of range	129	43 %
drift, of std	99	73 %
drift, of value	127	48 %
replay	21	5 %
step, of range	82	28 %
step, of std	65	78 %
step, of value	76	38 %

Table 15: How many attacks changed the decision, and how many of those were issued as TRUSTED.

de-weighting toward a reference that is itself off by about 1 standard deviation can push a decision near a level boundary. This is reported, not tuned away.

7.8 Blind rehearsal

Not done yet. It needs one person to inject at a site and interval the others do not know, and the work so far was done by one person. The file-mode tool (Section 10) is built for it, and it is the first planned rehearsal.

8 Key metrics at a glance

- **Detection quality.** Replay 100 %. Steps 70 / 90 / 35 % and drifts 81 / 94 / 49 % (of value / range / standard deviation) for detector S. Per-day ROC-AUC 0.728, TPR 19 % at 1 % FPR. Onset error 2–6 days for steps. Probability calibration Brier 0.131. Wrong-decision days −32 %.
- **False alarms.** S 4.7 and M 1.7 alarm events per station-year (1.3 % of days). Replay matcher 0. On extreme-weather days 2–12 % against 0.6–0.9 % on ordinary days. False quarantines in one of three station-years (Kansas, 29 days).
- **Latency.** Median days from onset to first alarm: replay 0, step 6 (range) to 12 (value), drift 30 to 40. To quarantine: replay 1, step 13–19. Processing 14.35 ms per reading including fits. Cadence: daily.
- **Memory.** Peak 29.5 MiB for 4 station-years. Per-station state 1,444 bytes + 23,521 bytes. Models 4.18 MiB (M) and 2,089 bytes (fusion).
- **Security overhead.** 118 bytes per record (139 bytes with 5 variables). Sign 62.4 µs, verify 150.5 µs, ledger append 2.4 µs.

9 Limits and what comes next

- **Small attacks are not caught.** Below about 0.7 standard deviations of the station-to-reference difference (every "12 % of the standard deviation" case, most wind cases). *Next:* a second independent witness: Sentinel-2 L2A indicators (an arbiter when station and reference disagree) and neighbouring stations, plus physical-consistency checks between variables.
- **Decisions can still flip early,** before the evidence is strong enough to act. With daily data and a 12 % change this floor cannot be removed without false alarms on honest stations. *Next:* margin-to-harm analysis per decision level.
- **Extreme weather is flagged somewhat more often than ordinary days.** *Next:* regime-dependent spreads.

– **Detector M is site-bound,** and its replay features include the scenario offsets. *Next:* site-independent calibration; replay features independent of the offsets.
– **Small evaluation base:** three sites, one test year, no held-out site in the detection tests, no blind rehearsal. *Next:* two held-out sites and a blind rehearsal with the file-mode tool.
– **File mode needs a clean start** (baseline poisoning, E7). *Next:* cold-start calibration from NASA POWER climatology and the training years of neighbouring stations.
– **Not evaluated:** F4–F9 as separate injections, and the evasions E3, E4, E8.

10 Reproducing the results

– `python -m pytest`: 164 unit and integration tests (1 min).
– `python scripts/run_all.py`: standard scenario, 756 attacked runs, detectors S and M (the tables of Sections 7.1 to 7.4; 10 min).
– `python scripts/run_trust.py`: probability, trust, responses, decisions, leave one site out (12 min).
– `python scripts/report_metrics.py`: layer 1 positions, cost, ROC and ablation, extreme weather, evasion sweeps (15 min).
– `python scripts/train_fusion.py`: the frozen fusion model used on new data (12 min).
– `python scripts/build_report.py`: the original report (HTML and PDF) from the results (1 min).

Every setting lives in one YAML file per stage (`config/`). Random seeds are fixed (42). Package versions are pinned in `requirements.txt` (numpy 2.5.4, pandas 3.0.6, scipy 1.18.1, scikit-learn 1.9.1). Hardware: laptop CPU only, no GPU.

10.1 File mode: any station series plus coordinates

This is what a test on a new site looks like. Run `python scripts/assess.py station.csv --lat ... --lon ...`. Columns are recognised automatically (NASA POWER names too) and sub-daily data is aggregated. NASA POWER for the place and days is downloaded once and cached. The calibration and detectors are fitted on the first part of the series (assumed clean; two thirds by default, or given dates) and the rest is scored.

Per day, the output is: detector scores, compromise probability, trust and band, the series after the responses, the decision and its confidence. It also lists every action with its reason and every episode with its estimated onset.

10.2 The control room

A graphical tool (`mingw32-make gui`) runs and shows everything:

– **Assess data:** file mode on any CSV or Excel file.
– **Simulate attack:** choose the site, attack, size, meaning of "12 %", onset, variables and attacker position P1/P2/P3, and get a verdict.
– **Experiments:** the commands above, with a live log and their tables.
– **Configs and docs.**

A Datasets

- **NASA POWER Daily API** (the reference). Provider: NASA Langley Research Center, https://power.larc.nasa.gov. Content: T2M, RH2M, WS2M, PRECTOTCORR, ALLSKY_SFC_SW_DWN at 3 points, 2021–2024. Licence: free; acknowledge the NASA Langley POWER Project. Responses cached with SHA-256 manifest.
- **AZMET daily.** Provider: University of Arizona, https://azmet.arizona.edu. Content: station az09: temperature, humidity, wind (3 m), rain, solar. Licence: free; cite the network and the access date.
- **FAWN 15-minute.** Provider: University of Florida IFAS, https://fawn.ifas.ufl.edu. Content: station Okahumpka (303), aggregated to daily. Licence: free to the public; yearly CSV files with SHA-256 in the repository.
- **Kansas Mesonet daily.** Provider: Kansas State University, https://mesonet.k-state.edu. Content: station Manhattan; rain = larger of two gauges. Licence: free for research with attribution (policy to confirm).
- **Copernicus Sentinel-2 L2A.** Provider: ESA / Copernicus Data Space, https://dataspace.copernicus.eu. Content: cloud, vegetation and moisture indicators. Licence: free, open; not used yet.

Cleaned files read by the pipeline (one schema, full daily calendar), committed in the repository, with the first 16 hex characters of their SHA-256:

File	Size	SHA-256
azmet_daily_clean.csv	50 KB	39658ac6e085f787...
fawn_daily_clean.csv	61 KB	ec09fcab7d55e858...
kansas_daily_clean.csv	56 KB	f6637a841ecedad1...
power_28.682_-81.886_daily_clean.csv	56 KB	81435b8a8bf80129...
power_32.48_-109.93_daily_clean.csv	55 KB	b7ea93b32352a17f...
power_39.209_-96.592_daily_clean.csv	55 KB	a219d8bf1a776b73...

Scripts re-download the raw data (`data/scripts/fetch_2021_2024.py`) and rebuild these files (`data/scripts/clean.py`). Known limits: rain agrees least with NASA POWER at every site (point against grid average), and there is one station per site, so the offsets are not valid for a whole climate.

B Parameters

- **Split** (`config/split.yaml`): train 2021-01-01 to 2023-12-31; test 2024.
- **Calibration** (`config/preparation.yaml`): per season (DJF, MAM, JJA, SON); median offset, MAD spread × 1.4826; log ratio for wind and rain with pseudocount 0.1; rain dry-dry days missing (≤ 0.1 mm).
- **CUSUM** (`config/detector_s.yaml`): AR(1) whitening; slack $k = 0.5$ standard deviations; threshold automatic (≤ 1 alarm per year on training); restart after alarm; episodes merged within 14 days.
- **Mahalanobis** (`config/detector_s.yaml`): robust centre and scale; shrinkage 0.10; χ^2 quantile 0.999, 4 degrees of freedom.
- **Replay matcher** (`config/detector_s.yaml`): 7-day windows; lags 1–730 days; tolerance 0.05 standard deviations.
- **Trend** (`config/detector_s.yaml`): 150-day window; whitened; threshold max(Bonferroni z at $\alpha = 0.001$, ≤ 1 alarm per year).

- **Detector M** (`config/config_m.yaml`): LOF (20 neighbours), contamination 0.01; 99th percentile of leave-one-year-out scores; granularity chosen on held-out years.
- **Fusion** (`config/trust.yaml`): logistic regression, $C = 1$, 9 features, persistence window 30 days.
- **Humidity witness** (`config/preparation.yaml`): ridge regression on 15 POWER features, $\alpha = 10$.
- **Trust** (`config/trust.yaml`): fall rate 0.5, recovery rate 0.05; de-weight below the 5 % clean quantile, quarantine below the 0.5 % one.
- **Responses** (`config/trust.yaml`): implicated = 14-day mean residual beyond 1 standard deviation; verification request at most every 14 days.
- **Product** (`config/trust.yaml`): 30-day water balance; dry below the 30th, severe below the 10th percentile per site and month.

Glossary

Residual How far the station is from the reference on a given day, after removing the usual difference. Measured in standard deviations.

Standard deviation A typical size of the normal day-to-day scatter. "2 standard deviations" means twice the usual scatter.

NASA POWER A public dataset of weather estimates from satellites and models, on a grid of about 50 km. Used here as the independent witness.

Witness / reference An independent source we compare the station with. An attacker at the station cannot change it.

Step bias False values shifted by a fixed amount from one day on.

Drift False offset that grows slowly day after day.

Replay Old, genuine data played back as if it were new.

Ed25519 A standard digital-signature method. Only the holder of the private key can sign, but anyone with the public key can check.

SHA-256 / hash A fingerprint of data. Changing one bit changes the whole fingerprint.

Hash chain / ledger Records linked so each one contains the fingerprint of the previous one. Editing the past breaks the chain.

CUSUM "Cumulative sum". Adds up small daily disagreements until they are too large to be chance.

Mahalanobis distance Measures how unusual a day is across several variables at once.

Mann-Kendall trend test A test for whether a series is steadily rising or falling.

LOF (local outlier factor) A learned method that flags data points that sit in an unusually sparse area compared with their neighbours.

One-class model A model trained only on normal data, which flags anything that looks different.

Logistic regression A simple model that turns several numbers into a probability.

Brier score Average squared error of predicted probabilities. Lower is better; 0.162 here is what "always guess the base rate" scores.

ROC-AUC How well a score separates attacked days from clean days. 0.5 is a coin flip, 1.0 is perfect.

TPR @ 1 % FPR Share of attacked days caught while only 1 % of clean days are falsely flagged.

False alarm An alert on clean data.

Latency Days between the start of an attack and the first alarm.

Quarantine The station's affected variables are replaced by the calibrated reference until trust returns.

ET0 Reference evapotranspiration: how much water a standard crop would lose to the air in a day.

P1, P2, P3 Attacker positions: at the station, on the link, and in the processing pipeline.

References

[1] Y. Liu, P. Ning, M. K. Reiter, "False data injection attacks against state estimation in electric power grids", ACM CCS 2009. doi:10.1145/1653662.1653666

[2] R. Langner, "Stuxnet: dissecting a cyberwarfare weapon", IEEE Security & Privacy 9(3), 2011. doi:10.1109/MSP.2011.67

[3] T. Trippel et al., "WALNUT: waging doubt on the integrity of MEMS accelerometers with acoustic injection attacks", IEEE EuroS&P 2017. doi:10.1109/EuroSP.2017.42

[4] D. F. Kune et al., "Ghost Talk: mitigating EMI signal injection attacks against analog sensors", IEEE S&P 2013. doi:10.1109/SP.2013.20

[5] World Meteorological Organization, Guide to Instruments and Methods of Observation (WMO-No. 8). https://community.wmo.int/node/37733

[6] R. G. Allen, L. S. Pereira, D. Raes, M. Smith, Crop evapotranspiration, FAO Irrigation and Drainage Paper 56, 1998. https://www.fao.org/4/x0490e/x0490e00.htm

[7] E. S. Page, "Continuous inspection schemes", Biometrika 41, 1954 (CUSUM). doi:10.1093/biomet/41.1-2.100

[8] H. B. Mann, "Nonparametric tests against trend", Econometrica 13, 1945 (doi:10.2307/1907187); M. G. Kendall, Rank Correlation Methods, 1975.

[9] M. M. Breunig et al., "LOF: identifying density-based local outliers", ACM SIGMOD 2000. doi:10.1145/335191.335388

[10] D. J. Bernstein et al., "High-speed high-security signatures" (Ed25519), Journal of Cryptographic Engineering 2, 2012. doi:10.1007/s13389-012-0027-1

[11] NASA Langley Research Center, POWER Project, Daily API. https://power.larc.nasa.gov